QDA

- Liad Mandil, 315096073
- Maxim Prokopchuk, 337875397

Imports

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_curve, auc, RocCurveDisplay
from sklearn.pipeline import Pipeline
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score, ConfusionMatrixDisplay
from sklearn.discriminant_analysis import QuadraticDiscriminantAnalysis as QDA
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.model_selection import cross_val_score
```

Helpfull code

Load Data

In [2]:
```
spaceship_data = pd.read_csv('final_data.csv')
```

split to x and y

In [3]:
```
y = spaceship_data["Transported"].astype(int)
X = spaceship_data.drop(columns=["Transported", "GroupId"])
```

Train-Test Split

In [4]:
```
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

In [5]:
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

QDA Model

In [6]:
```
qda_pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("qda", QDA())
])

param_grid = {
    "qda__reg_param": np.linspace(0.0, 0.2, 20)
}

qda_grid = GridSearchCV(
    qda_pipe,
    param_grid,
    cv=cv,
    scoring="accuracy",
    n_jobs=-1
)

qda_grid.fit(x_train, y_train)

print("Best QDA params:", qda_grid.best_params_)
print(f"Best CV accuracy: {100 * qda_grid.best_score_:.2f}%")

best_model = qda_grid.best_estimator_
```

```
Best QDA params: {'qda__reg_param': np.float64(0.0)}
Best CV accuracy: 75.80%
```

```
c:\Users\Liad M\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn
\discriminant_analysis.py:1024: LinAlgWarning: The covariance matrix of class 0 i
s not full rank. Increasing the value of parameter `reg_param` might help reducin
g the collinearity.
  warnings.warn(
c:\Users\Liad M\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn
\discriminant_analysis.py:1024: LinAlgWarning: The covariance matrix of class 1 i
s not full rank. Increasing the value of parameter `reg_param` might help reducin
g the collinearity.
  warnings.warn(
```

In [7]:
```
results_df = pd.DataFrame(qda_grid.cv_results_)

# עמודות חשובות בלבד
cols = [
    "rank_test_score",
    "mean_test_score",
    "std_test_score",
    "param_qda__reg_param"
]

top5 = (
    results_df[cols]
    .sort_values(by="mean_test_score", ascending=False)
    .head(5)
    .reset_index(drop=True)
)

# באחוזים נוחים לקריאה
top5["mean_test_score"] = top5["mean_test_score"] * 100
```

```
top5["std_test_score"]  = top5["std_test_score"] * 100

top5
```

Out[7]:

	rank_test_score	mean_test_score	std_test_score	param_qda__reg_param
0	1	75.799715	1.613228	0.000000
1	2	73.448273	1.640098	0.200000
2	3	73.243787	1.603521	0.189474
3	4	73.024692	1.673571	0.178947
4	5	72.966276	1.674649	0.168421

Results

In [8]:
```
best_model = qda_grid.best_estimator_
```

In [9]:
```
y_pred_qda_train  = best_model.predict(x_train)
y_score_qda_train = best_model.predict_proba(x_train)[:, 1]
```

In [10]:
```
train_acc_qda = accuracy_score(y_train, y_pred_qda_train)
```

In [11]:
```
print("="*40)
print("BEST QDA — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_qda*100:.2f}%")
print("="*40)
```

```
========================================
BEST QDA — TRAIN SET PERFORMANCE
========================================
Accuracy : 74.24%
========================================
```

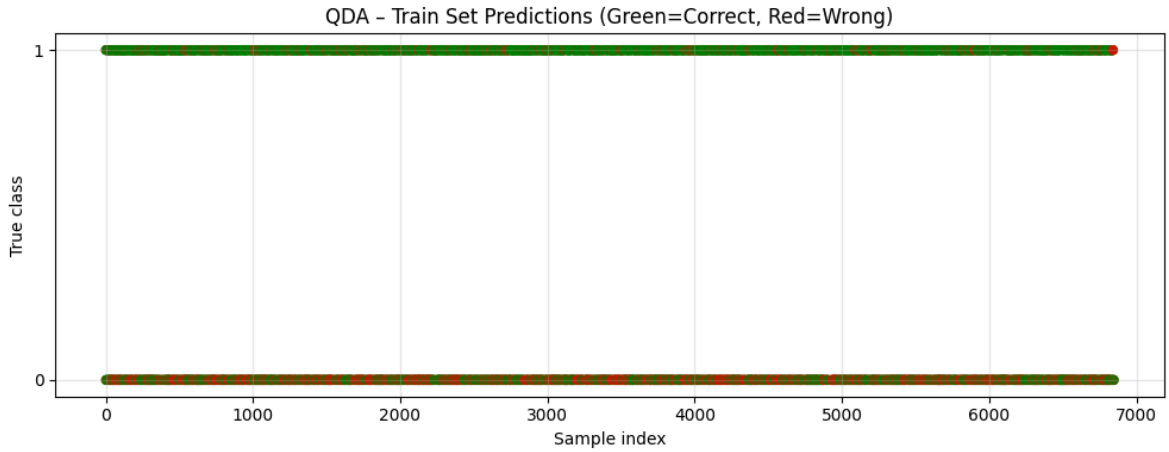
מטריצת ערבול למודל המוצלח ביותר TRAIN

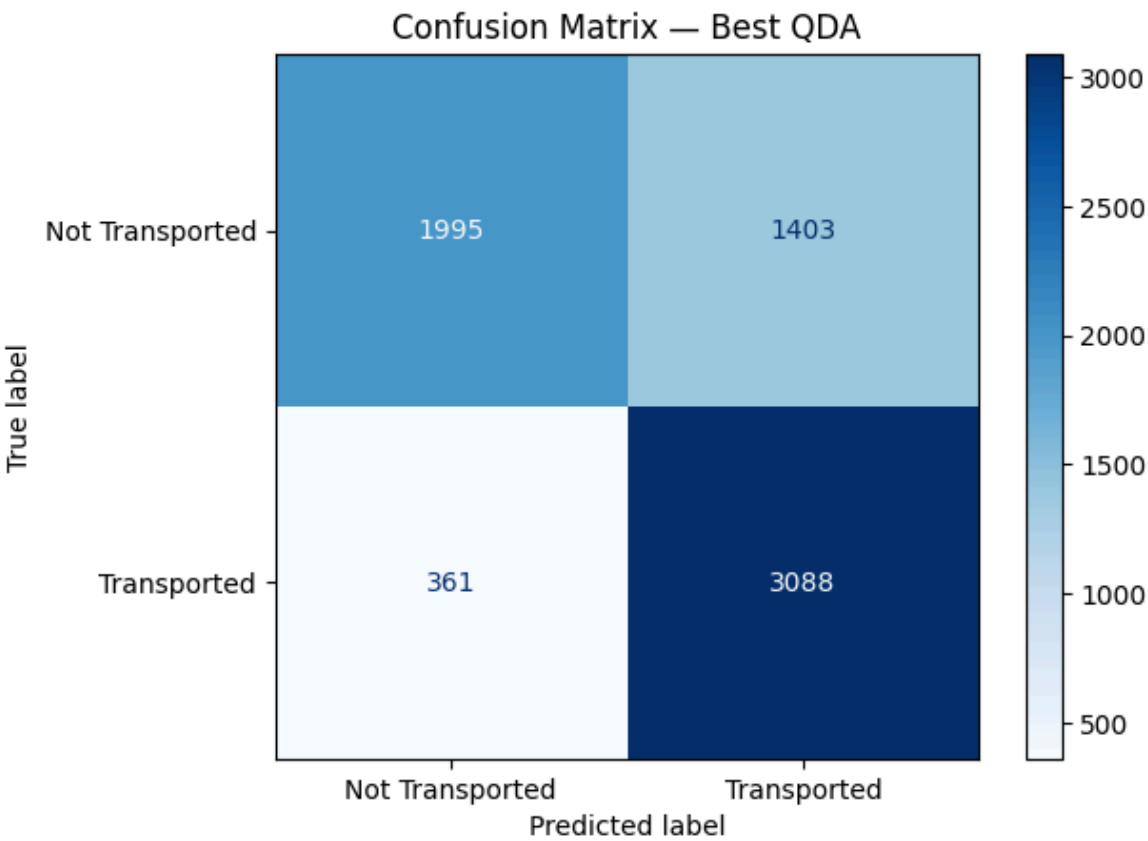
In [12]:
```
correct = (y_pred_qda_train == y_train)

colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_train)), y_train, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("QDA – Train Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
Errors: 1764 / 6847
```

In [13]:
```python
ConfusionMatrixDisplay.from_estimator(
    best_model,
    x_train,
    y_train,
    display_labels=["Not Transported", "Transported"],
    cmap="Blues"
)

plt.title("Confusion Matrix — Best QDA")
plt.grid(False)
plt.show()
```